Project Description

Scientific Calculator using Java Swing

Overview

This project is a scientific calculator for the desktop, written in Java. It opens in its own window with a large black display at the top and a grid of dark buttons underneath. You click the buttons to enter numbers and operations, and the answer appears on the display.

Besides the usual add, subtract, multiply and divide, it can work out powers, roots, logarithms, factorials, and trigonometric and hyperbolic functions. It follows the normal order of operations, so *2 + 3 * 4* gives 14, and it understands brackets.

Purpose

The aim was to build a calculator that is useful for students, who often need more than a basic calculator offers, and at the same time to practise the main ideas of GUI programming in Java: layouts, buttons, event handling and turning typed input into a correct result.

Key Features

- Basic arithmetic with decimal numbers, brackets and percentage
- Correct order of operations (BODMAS)
- Powers and roots: x^2, x^3, y^x, 2^x, square root and cube root
- Logarithm (base 10), reciprocal (1/x), factorial (x!) and sign change (+/-)
- Trigonometric functions (sin, cos, tan) and hyperbolic functions (sinh, cosh, tanh)
- Rad/Deg switch for the angle mode
- Buttons for pi and a random number
- Shows "Error" instead of crashing when the input is invalid, for example when dividing by zero

User Interface

The window is 1000 x 720 pixels and uses a dark theme that is easy on the eyes. The display shows the numbers right-aligned in a bold light-blue font. The buttons are laid out in an 8-column, 6-row grid. The **C** and **=** buttons are red so they are easy to find, the **=** button is two rows tall, and the **0** button is three columns wide, just like on a real calculator.

How It Works

Every button click is sent to one event-handling method. Digits and operators are added to a text expression on the display. Function buttons such as sin or square root act on the number currently shown and replace it with the answer. When **=** is pressed, a small expression evaluator reads the text and solves it in order: brackets and numbers first, then powers, then multiplication and division, and finally addition and subtraction.

Technical Details

Item	Details
Language	Java (JDK 8 or higher)
GUI toolkit	Swing and AWT (included with Java)
Layouts used	BorderLayout and GridBagLayout
Code structure	A single file, ScientificCalculator.java
External libraries	None
Platform	Windows, macOS and Linux

How to Run

Save the code as **ScientificCalculator.java**, open a terminal in the same folder, and run `javac ScientificCalculator.java` followed by `java ScientificCalculator`.

Limitations and Future Scope

The current version has no backspace button, no calculation history and no keyboard input. These can be added later, along with inverse trigonometric functions, natural logarithm, memory buttons and a light theme.